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ADC - Homework #1

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Part I

10303

4.2-6:

Band limited at 5 kHz
carrier 20 kHz

first carrier frequency: $f_1 = \frac{20000}{2} = 10000 \text{ Hz}$

second carrier frequency: $f_2 = \frac{40000}{2} = 20 \text{ kHz}$

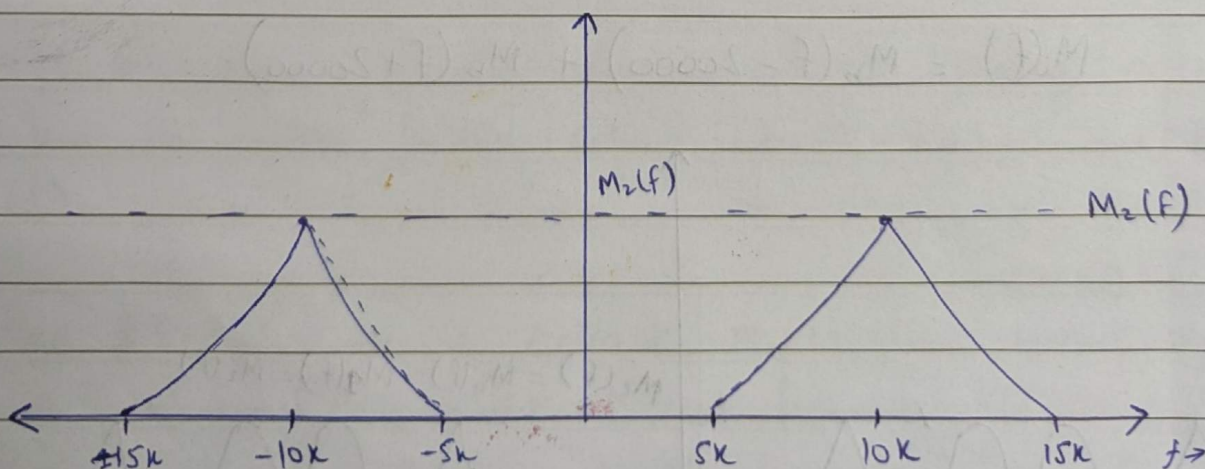
a) AT point a:

$$m_a(t) = m_2(t) \cdot 2\cos(20\pi t)$$

fourier transform:

$$M_a(f) = M_2(f - 10000) + M_2(f + 10000)$$

∴ as cosin function is used, we will observe two peaks on the plot.

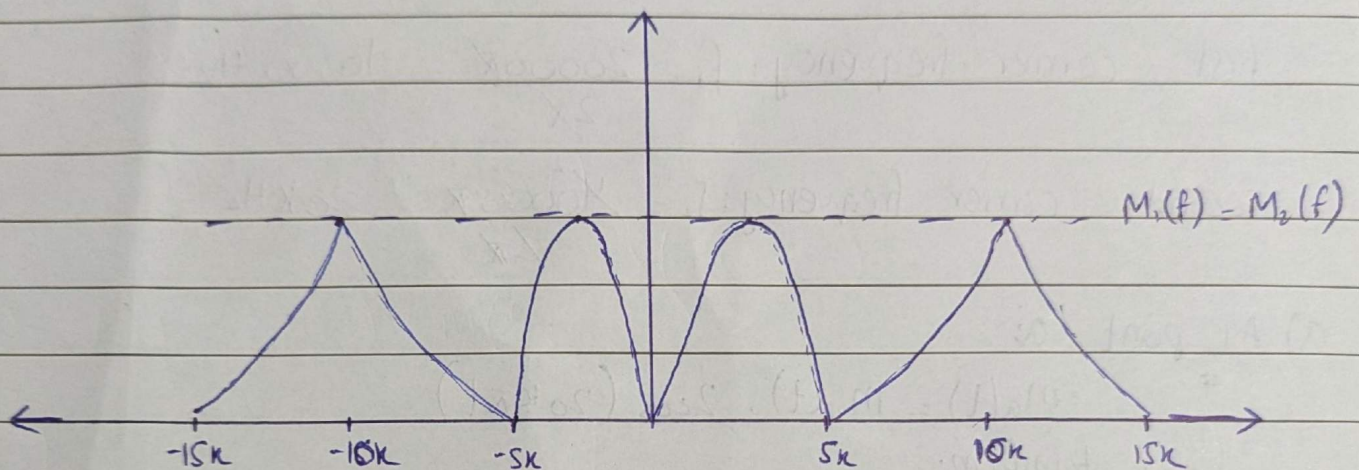


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at point b:

$$m_b(t) = m_1(t) + m_2(t)$$

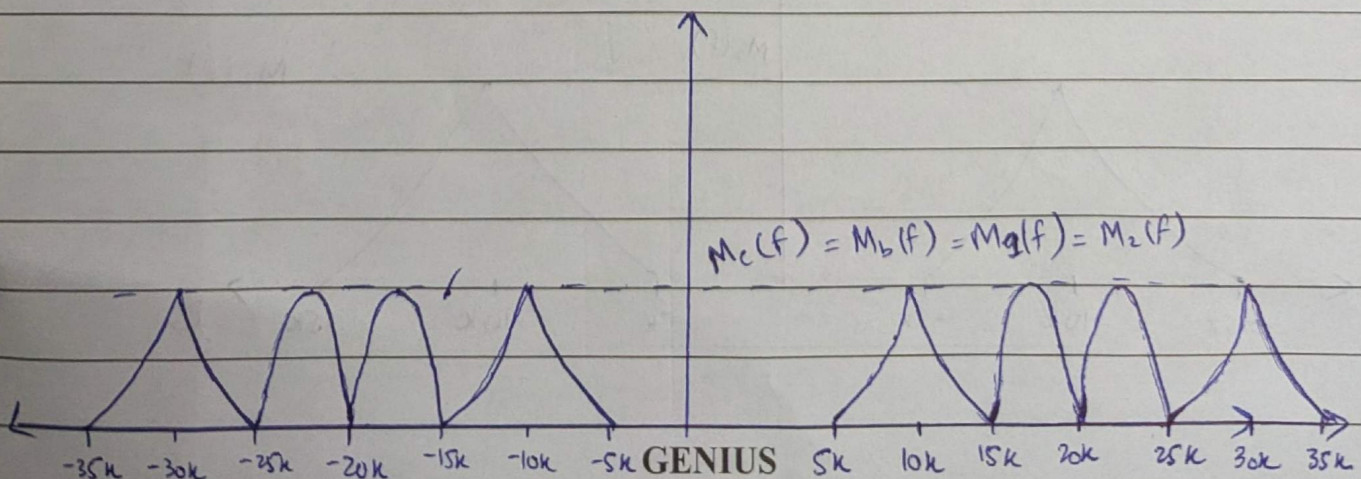
$$M_b(f) = M_1(f) + M_2(f)$$



at point c:

$$m_c(t) = m_b(t) \cdot \cos(40000\pi t) \times 2$$

$$M_c(f) = M_b(f - 20000) + M_b(f + 20000)$$



b) channel Bandwidth:

as the signal last from 5kHz to 35kHz so the channel bandwidth will be:

$$35000 - 5000 = 30,000 \text{ Hz.}$$

$$\text{Bandwidth} = 30 \text{ kHz}$$

c) Receiver Design:

as we know that at the receiver end we again multiply the modulated signal with the carrier signal to receive the message signal so:

- Stage #1: Multiply ~~the~~ $m_c(t)$ by $\cos(40000\pi t)$
now pass through the lowpass filter of cutoff at 15kHz.
now the signal received is $M_b(f)$.
- Stage #2: Now we extract $m_1(t)$ by passing $m_b(t)$ through a low pass filter with cutoff at 5kHz so output is only $m_1(t)$.
- Stage #3: Now ~~we~~ we extract $m_2(t)$ by passing $m_b(t)$ through a band pass filter of ~~the~~ frequencies between 5kHz to 15kHz, output will be $M_a(t)$. to get $m_2(t)$ we multiply it by $\cos(20000\pi t)$.

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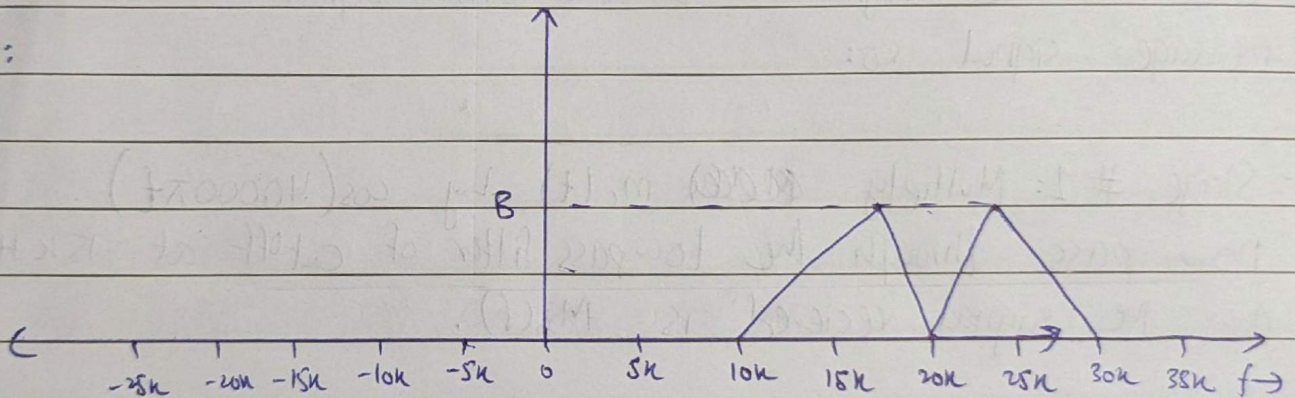
4.2-7:

a) $\omega_0 = 20,000\pi$

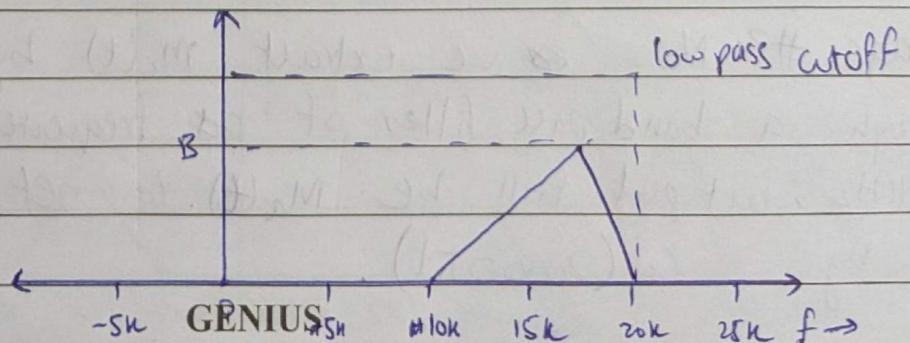
$$f = \frac{20000\pi}{2\pi} = 10\text{kHz}$$

$$x(t) = m(t) \cdot 2\cos(40000\pi t)$$

$$X(f) = M(f - 20000) + M(f + 20000)$$

 $M(f)$:Spectrum of $x(t)$:-

$X(f)$: as low pass will only pass below 20kHz so half the spectra will be observed



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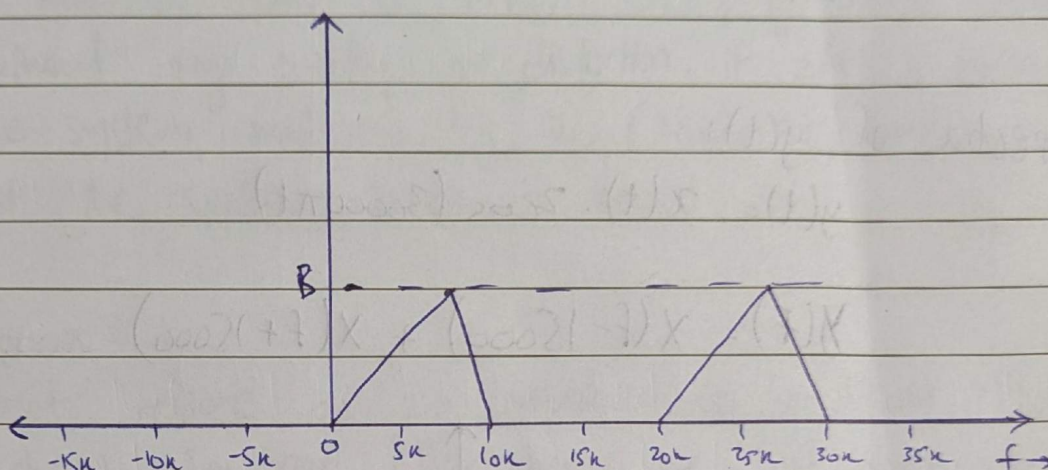
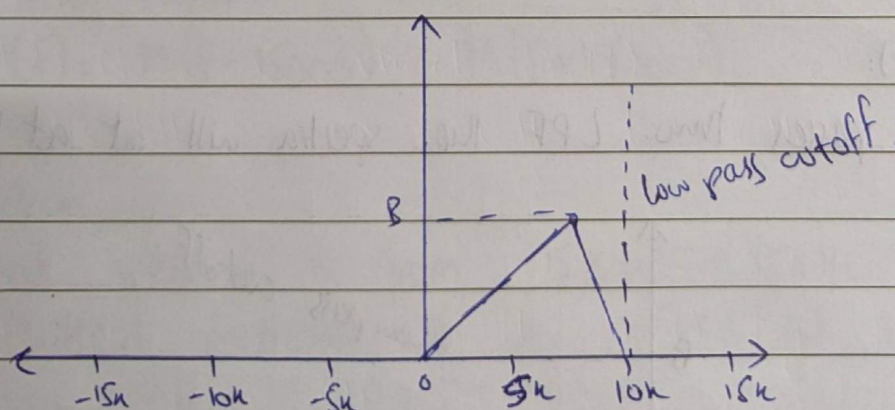
Spectra of $y(t)$:

$$y(t) = x(t) \cdot 2\cos(20000\pi t)$$

$$\therefore \frac{20000\pi}{2\pi} = 10000 \text{ Hz}$$

$$Y(f) = X(f-10000) + X(f+10000)$$

So,

• Spectra of $z(t)$:as the output of LPF (0-10kHz) is $z(t)$ so: $y(f) \rightarrow \text{LPF (0-10kHz)} \rightarrow z(f)$ here:-

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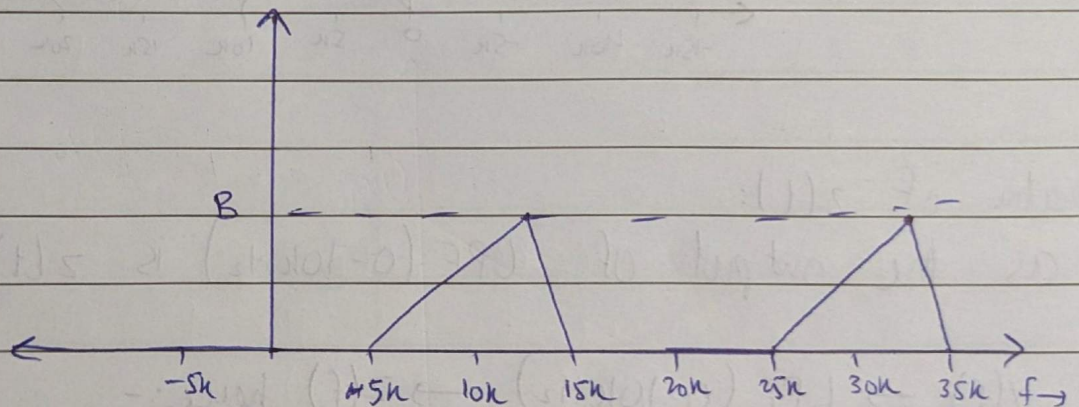
b) Spectra of $y(t)$ and $z(t)$ if $\omega_0 = 30000\pi$

$$f = \frac{30000\pi}{2\pi} = 15\text{ kHz}$$

Spectra of $y(t)$:

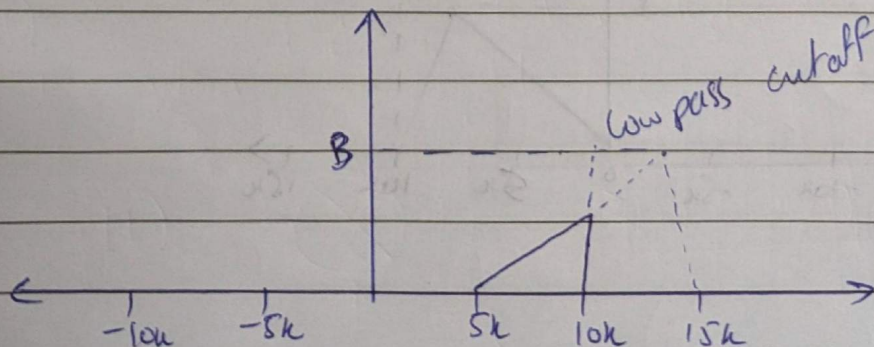
$$y(t) = x(t) \cdot 2\cos(30000\pi t)$$

$$Y(f) = X(f - 15000) + X(f + 15000)$$



Spectra of $z(t)$:

as $y(t)$ passes thru. LPF the spectra will be at 10 kHz.



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c)

• for $\omega_0 = 20,000\pi$:

Yes, we can descramble $z(t)$ to recover $m(t)$. as $z(t)$ is a spectral-inverted version of $m(t)$ occupying same 0-10kHz Band width without any overlap. or distortion. if $z(t)$ is passed through the same system, multiply by $2\cos(20000\pi t)$ and filter, will invert the spectrum back to $m(t)$.

• for $\omega_0 = 30,000\pi$:

No, we cannot restore as the negative side band at $\nu(f)$ is chopped due to low pass filter hence we lost some information that cannot be restored.

4.2-10:

given:

$$y(t) = m(t) \cdot 2 \cos(30000\pi t)$$

fourier transform:-

$$Y(f) = M(f - 15000) + M(f + 15000)$$

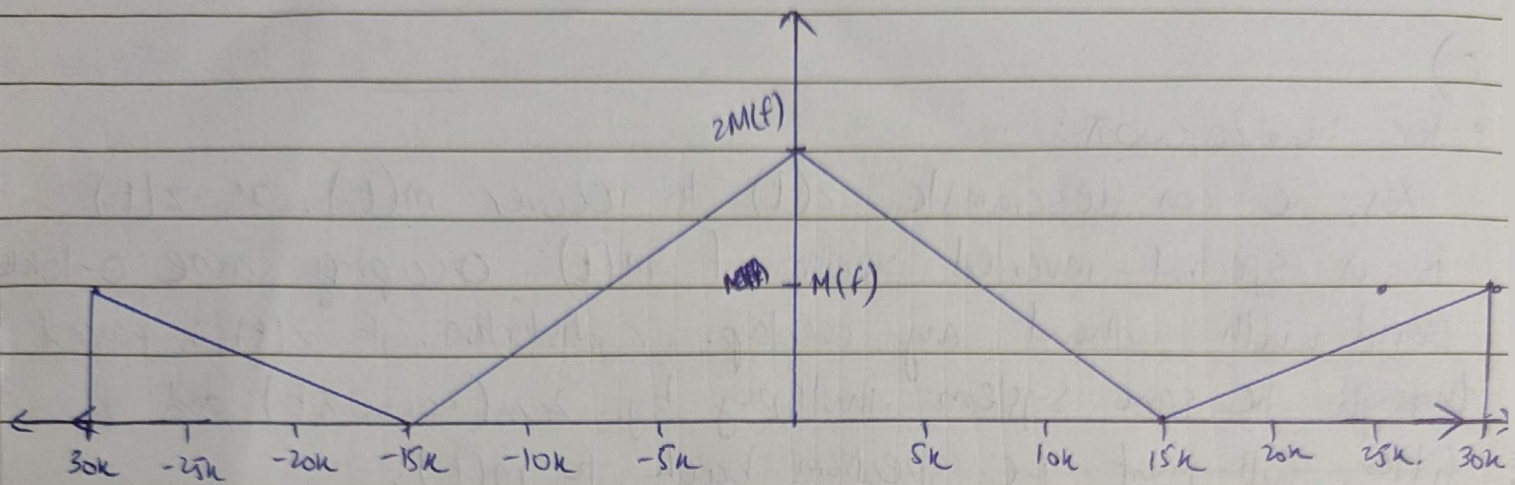
$$\therefore \frac{\omega_0}{2\pi} = f = \frac{30000\pi}{2\pi} = 15000 \text{ Hz}$$

15000 Hz

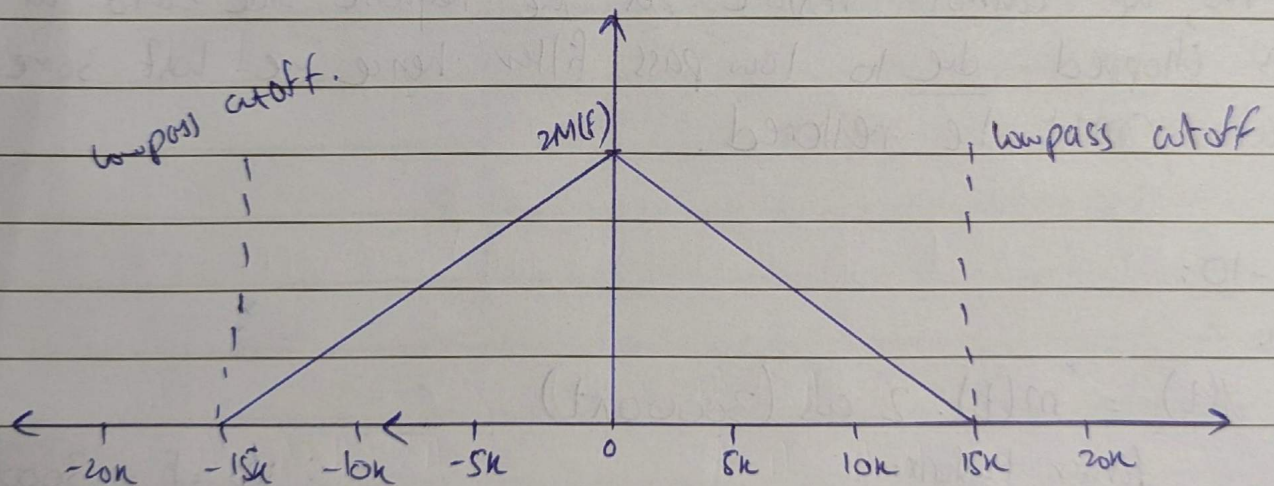
so, spectrum

a) original spectrum is from -15kHz to 15kHz.

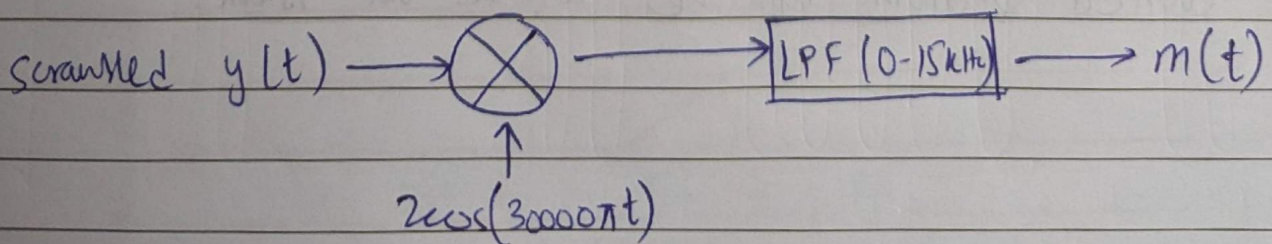
so shifted spectrum will be centered at 15kHz.



after passing through LPF (0-15 kHz) $Y(f)$:



b) Since $Y(f)$ occupies the exact same bandwidth (15 kHz) as $M(f)$ so simply spectral inverted, passing $y(t)$ through an identical system will invert the spectrum back to original.



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Problem 4.3-1:

as $m(t)$ has two binary levels at A and $-A$ so, we have.

$$s(t) = [B + m(t)] \cos \omega_c t$$

so,

$$m(t) = \begin{cases} +A \\ -A \end{cases}$$

so,

Am signal becomes.

$$s(t) = \begin{cases} [B + A] \cos \omega_c t & m(t) = +A \\ [B - A] \cos \omega_c t & m(t) = -A \end{cases}$$

only two carrier amplitudes are possible

$B + A$ and $B - A$

we will find B from modulation index:-

$$M = \frac{A}{B}$$

so

$$B = \frac{A}{M}$$

solving for each case:

$$M = 0.5$$

$$B = 2A$$

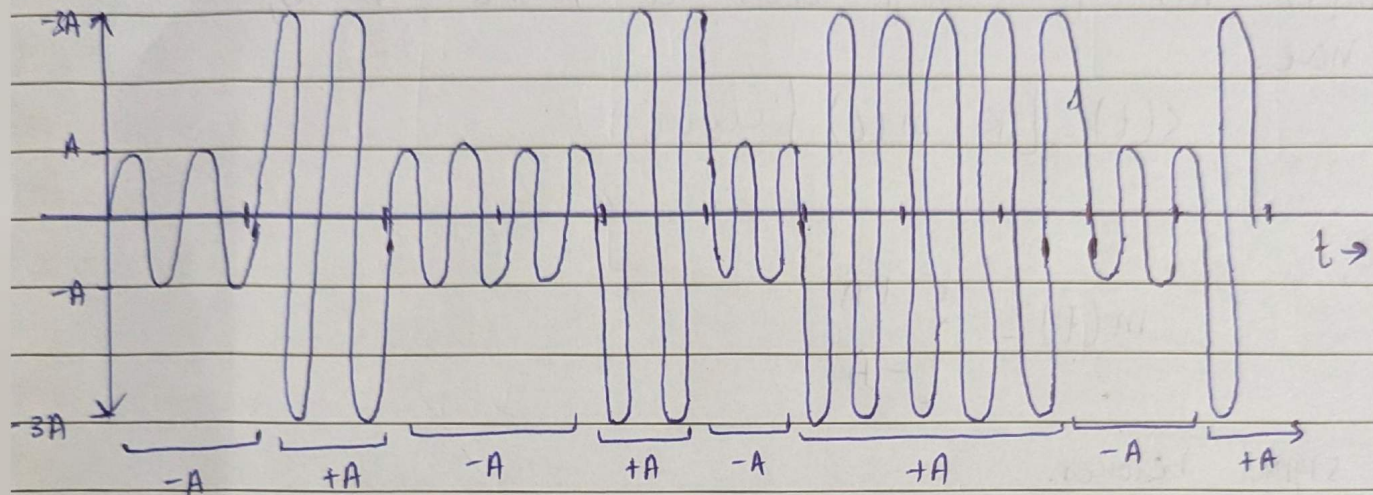
so

when $m = +A$ when $m = -A$:

$$B + A = 2A + A = 3A$$

$$B - A = 2A - A = A \quad \text{So carrier switches between } 3A \text{ and } A$$

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Sketch when $M=0.5$:Now when $M=1$:

$$\frac{B=A}{1} = A$$

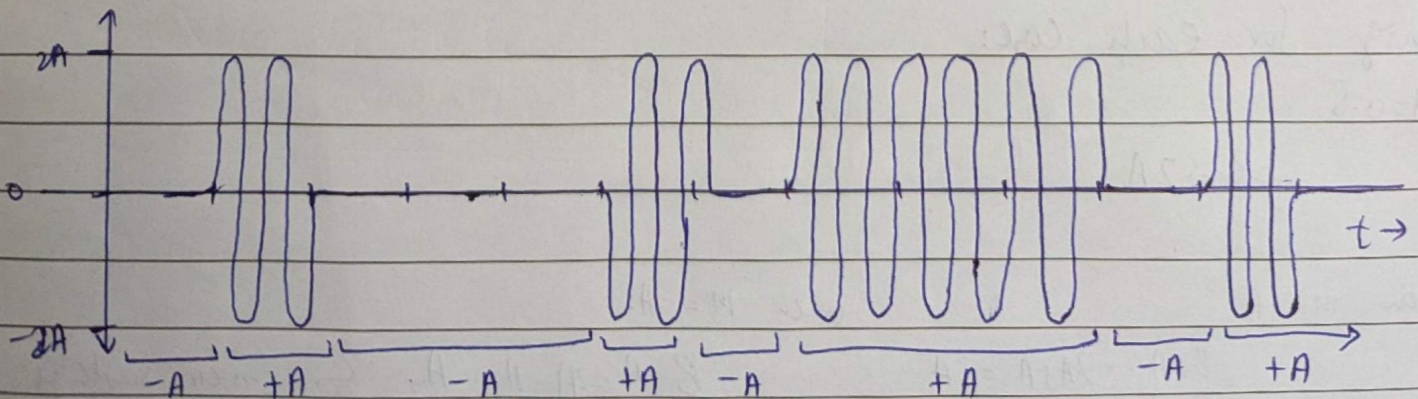
So,

when $m=+A$

$$B+A = A+A = 2A$$

when $m=-A$

$$B-A = A-A = 0$$

Sketch when $M=1$:

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c) $M=2$

when $m=+A$:

$$B+A = \frac{A}{2} + A = \frac{3}{2}A$$

when $M=-A$

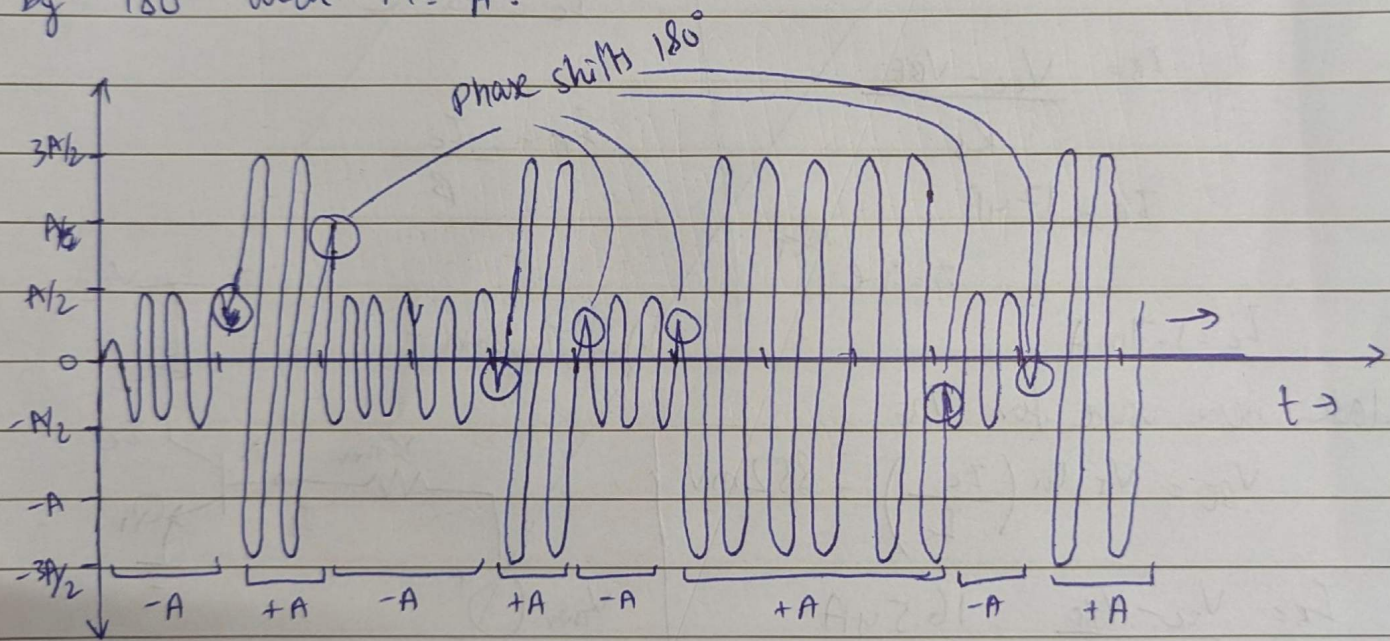
$$B-A = \frac{A}{2} - A = -\frac{A}{2}$$

$$S(t) = \begin{cases} \frac{3A}{2} \cos \omega_c t & m=+A \\ -\frac{A}{2} \cos \omega_c t & m=-A \end{cases}$$

so,

$$-\frac{A}{2} \cos \omega_c t = \frac{A}{2} \cos(\omega_c t + \pi) \text{ so carrier flips}$$

by 180° when $m=-A$.



d) $B = \frac{A}{4} = 0$

when $m=+A$

$$B+A = A$$

when $m=-A$

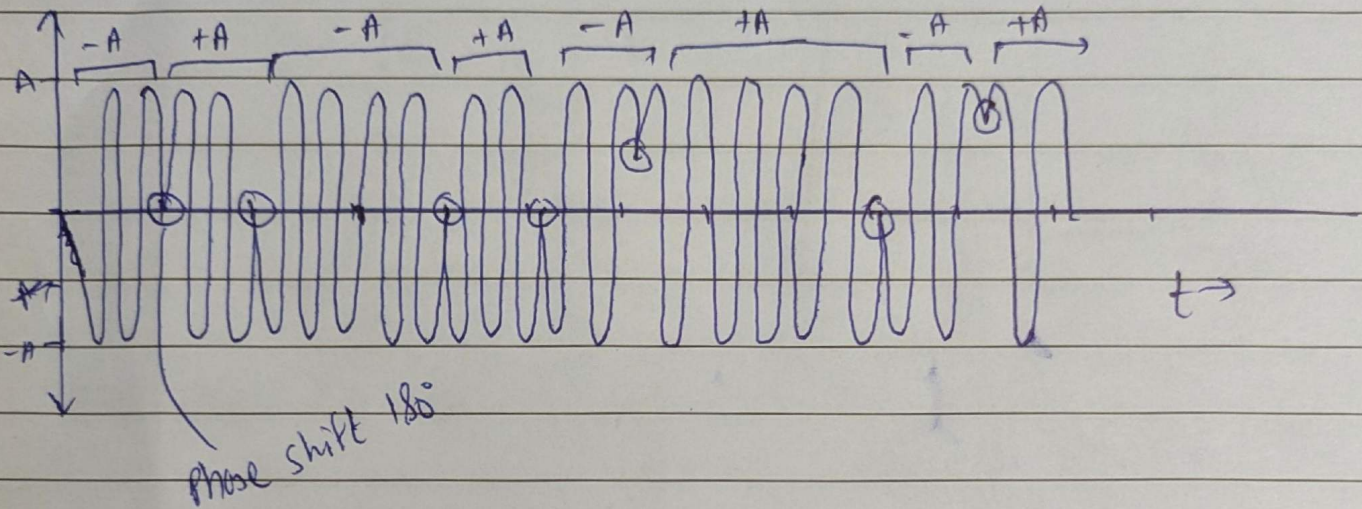
$$B+A = A$$

so amplitude remains constant whether $m=A$ or $-A$.

$$S(t) = \pm A \cos(\omega_c t) \text{ phase flips by } 180^\circ$$

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sketch:-



No, there is no separate carrier component for $\mu = \infty$ because we need $B \cos \omega t$ term, but here $B = 0$. There is no pure component for $\mu = \infty$

4.3-2:

carrier frequency = 2 kHz

$$\text{output} = s_{\text{am}}(t) = 2[b + 0.5m(t)] \cos \omega_c t$$

$$m(t) = \pm A$$

$$E[m(t)] = 0$$

$$E[m^2(t)] = A^2$$

a) Average power as a function of b and A :

power of $x(t) \cos \omega_c t$ is:

$$P = \frac{1}{2} E[x^2(t)]$$

So,

$$P_{\text{AM}} = \frac{1}{2} E[(2b + m)^2]$$

$$P_{\text{AM}} = \frac{1}{2} [4b^2 + 4bE[m] + E[m^2]]$$

put values:

$$P_{\text{AM}} = 2b^2 + \frac{A^2}{2}$$

$$\text{carrier power} = P_c = \frac{(2b)^2}{2} = 2b^2$$

side band modulation is:

$$P_{\text{SB}} = \frac{A^2}{2}$$

b) $b = A$

$$A_c = 2b = 2A$$

So,

$$M = \frac{A}{2A} = \boxed{0.5}$$

Carrier power:

$$P_c = \frac{(2A)^2}{2} = 2A^2$$

Side band:

$$P_{SB} = \frac{A^2}{2}$$

$$\text{Total} = P_T = 2A^2 + \frac{A^2}{2} = \frac{5}{2} A^2$$

Modulation efficiency is given by $\eta = \frac{P_{SB}}{P_T} = \frac{A^2/2}{5A^2/2} = \boxed{1/5}$

$$\boxed{\eta = 20\%}$$

c) for envelope detection, the AM envelope must never become negative.

$$2b + m(t) \geq 0$$

So,

$$m(t) = -A \quad (\text{worst case})$$

then

$$2b - A \geq 0$$

$$b \geq \frac{A}{2}$$

so

$$\boxed{b_{\min} = \frac{A}{2}}$$

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$$A_c = 2b = A$$

So,

$$\boxed{M_{\max} = \frac{A}{A} = 1} \quad \text{max modulation index} = 1$$

max modulation efficiency at $b = A/2$

$$P_c = 2 \left(\frac{A}{2} \right)^2 = \frac{A^2}{2}$$

$$\text{Side band power} = P_{SB} = \frac{A^2}{2}$$

So,

$$P_T = A^2$$

and,

$$\eta_{\max} = \frac{A^2/2}{A^2} = \frac{1}{2} = 0.5$$

$$\& \text{ that is } \boxed{50\%}$$

4.3-3:

$$T_m = 0.1 \text{ seconds}$$

$$f_m = \frac{1}{0.1} = 10 \text{ Hz}$$

$$\text{carrier} = 1000 \text{ Hz}$$

Case $M=0.5$:

$$M = \frac{A}{B}$$

$$B = 2A$$

So,

$$s(t) = [2A + m(t)] \cos \omega_c t$$

so varies from

$$\boxed{A \text{ to } 3A} \text{ as } m(t) \text{ goes } A \text{ to } -A$$

Case $M=1$:

$$B = A$$

$$s(t) = [A + m(t)] \cos \omega_c t$$

$$\text{amplitude} = \boxed{0 \text{ to } 2A} \text{ as } m(t) \text{ goes } A \text{ to } -A.$$

Case $M=2$:

$$B = \frac{A}{2}$$

$$s(t) = \left[\frac{A}{2} + m(t) \right] \cos \omega_c t$$

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carrier amplitude ranges from:

$$-A/2 \text{ to } 3A/2$$

↑

180° phase shift. (when $m(t) < -A/2$)

case $\mu = \infty$:

$$M = \frac{A}{B} \rightarrow \infty$$

$$B = 0$$

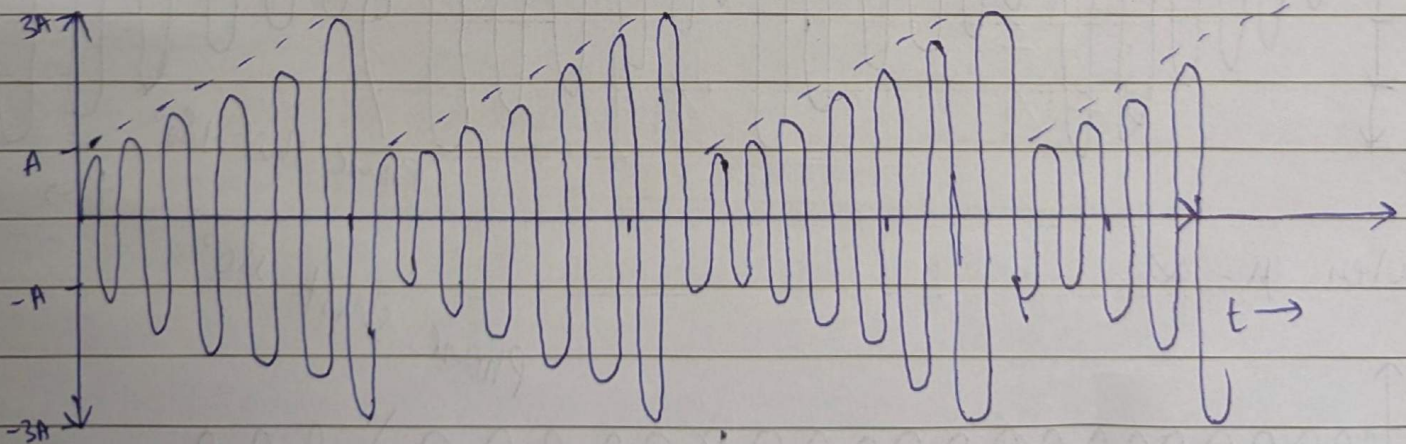
So,

$$s(t) = m(t) \cos \omega_c t$$

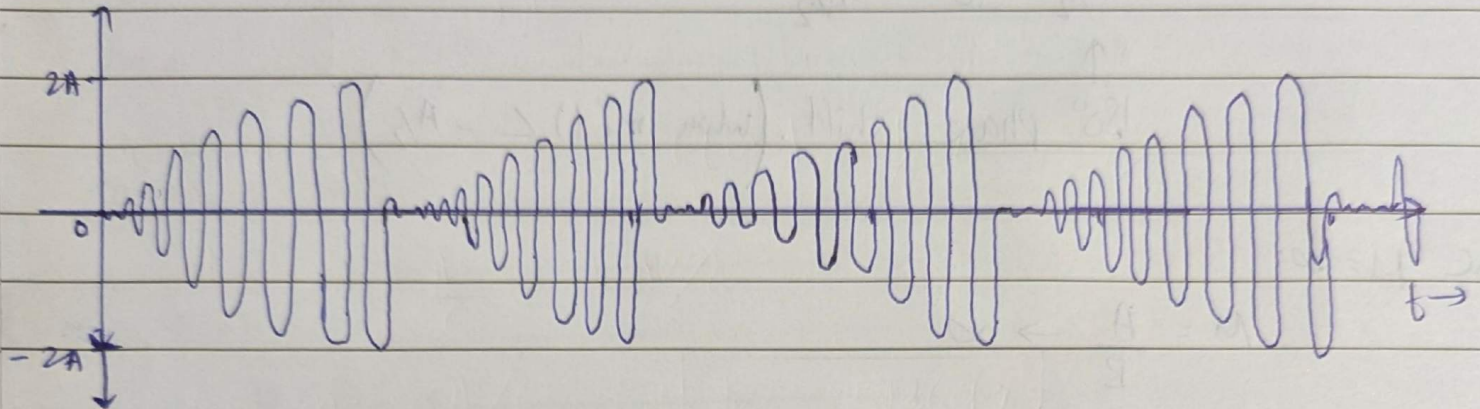
so there is no separate carrier term.

Sketches:-

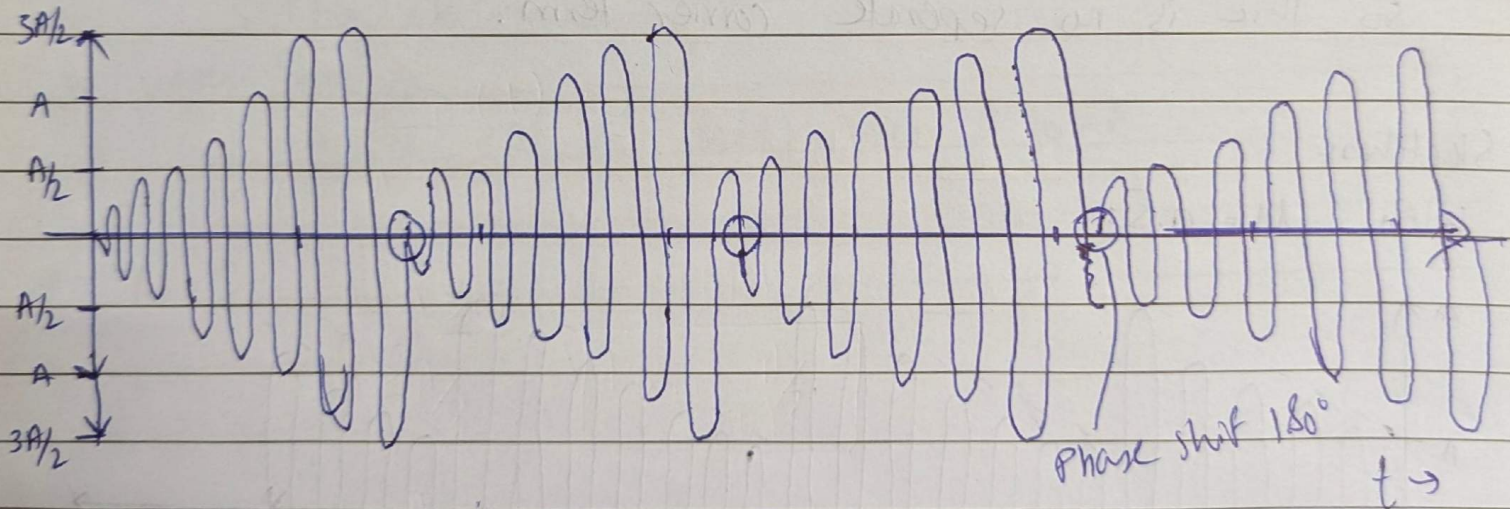
When $\mu = 0.5$:



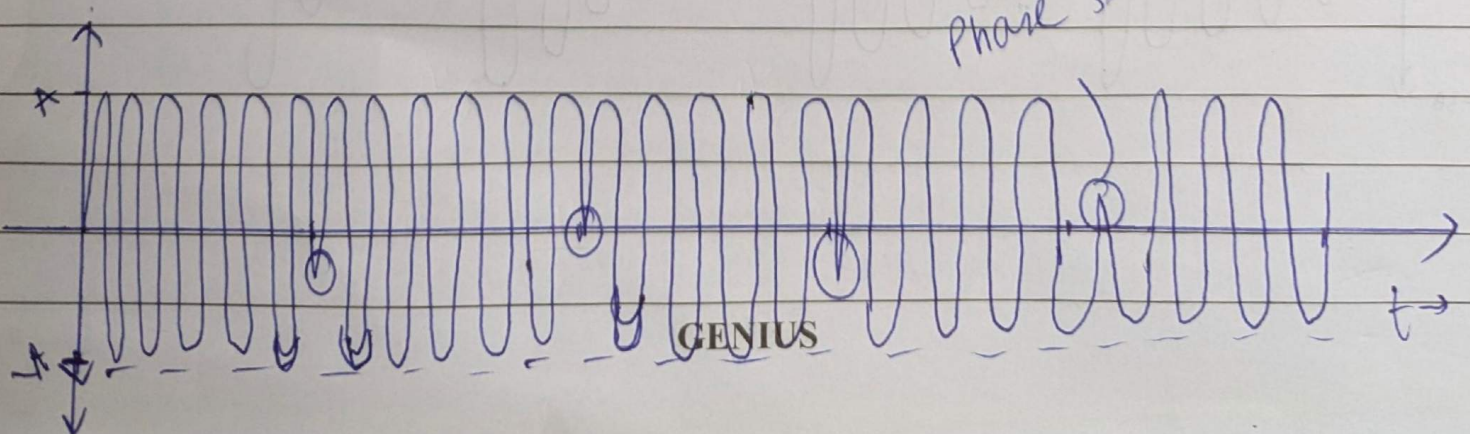
• when $M=1$:



• when $M=2$:



• when $\mu = \infty$



4.3-4:

as sawtooth sweeps uniformly from $-A$ to $+A$:

$$E[m] = 0$$

mean square:-

$$E[m^2] = \frac{1}{2A} \int_{-A}^A m^2 dm$$

$$E[m^2] = \frac{1}{2A} \left[\frac{m^3}{3} \right]_{-A}^A$$

$$E[m^2] = \frac{1}{2A} \frac{2A^3}{3} = \frac{A^2}{3}$$

$$\text{Signal} = s(t) = [2b + m(t)] \cos \omega_c t$$

a) Average power:

$$P_T = \frac{1}{2} E[(2b + m)^2]$$

$$\text{as } E[m] = 0$$

$$P_T = \frac{1}{2} (4b^2 + E[m^2]) = \boxed{2b^2 + \frac{A^2}{6}}$$

b) $b = A$:

$$\text{carrier amp} = A_c = 2A$$

$$\text{modulation index} = \mu = \frac{A}{2A} = \boxed{0.5}$$

• Carrier power:

$$P_c = 2A^2$$

$$\text{side band power} = P_{SB} = \frac{1}{2} E[m^2] = \frac{1}{2} \frac{A^2}{3} = \frac{A^2}{6}$$

$$\text{Total} = 2A^2 + \frac{A^2}{6} = \frac{13A^2}{6}$$

$$\text{efficiency} = \eta = \frac{\frac{A^2}{6}}{\frac{13A^2}{6}} = \boxed{\frac{1}{13} = 7.69\%}$$

c) Minimum b for envelope detection:

$$2b + m(t) > 0$$

$$\text{lowest } m = -A$$

$$2b - A > 0$$

$$b \geq \frac{A}{2}$$

$$\boxed{b_{\min} = \frac{A}{2}}$$

$$A_c = A$$

and

$$\boxed{\mu_{\max} = 1}$$

$$\text{Power} = P_c = \frac{A^2}{2} \quad P_{SB} = \frac{A^2}{6}$$

$$P_T = \frac{2A^2}{3}$$

$$\text{efficiency} = \frac{A^2/6}{2A^2/8} = \frac{1}{4} = 25\%$$

$$\boxed{\eta_{\max} = 25\%}$$

Problem 4.3-5:

it's a symmetric triangular wave:

$$-10 \leq m(t) \leq 10$$

The figure gives $A=10$

$$\mu = 2$$

so,

$$\mu = \frac{A}{A_c}$$

$$2 = \frac{10}{A_c} \quad \boxed{A_c = 5}$$

a) Carrier amplitude and power:

$$\boxed{A_c = 5}$$

$$\text{Power} = P_c = \frac{A_c^2}{2} = \frac{25}{2} = \boxed{12.5 \text{ W}}$$

b) side band efficiency

$$m_{\text{rms}} = \frac{10}{\sqrt{3}}$$

$$m^2_{\text{rms}} = \frac{100}{3}$$

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Side band power is given by $P_{SB} = \frac{1}{2} m^2_{rms}$

$$P_{SB} = \frac{100}{6} = \boxed{16.67 \text{ W}}$$

$$\begin{aligned} \text{Total power} = P_T &= P_c + P_{SB} \\ &= 12.5 + 16.67 = 29.17 \text{ W} \end{aligned}$$

$$\text{efficiency} = \eta = \frac{16.67}{29.17} \approx \boxed{57.13\%}$$

4.3-6:

As we know $A_c = 5$

therefore:-

$$S_{AM}(t) = [5 + m(t)] \cos \omega_c t$$

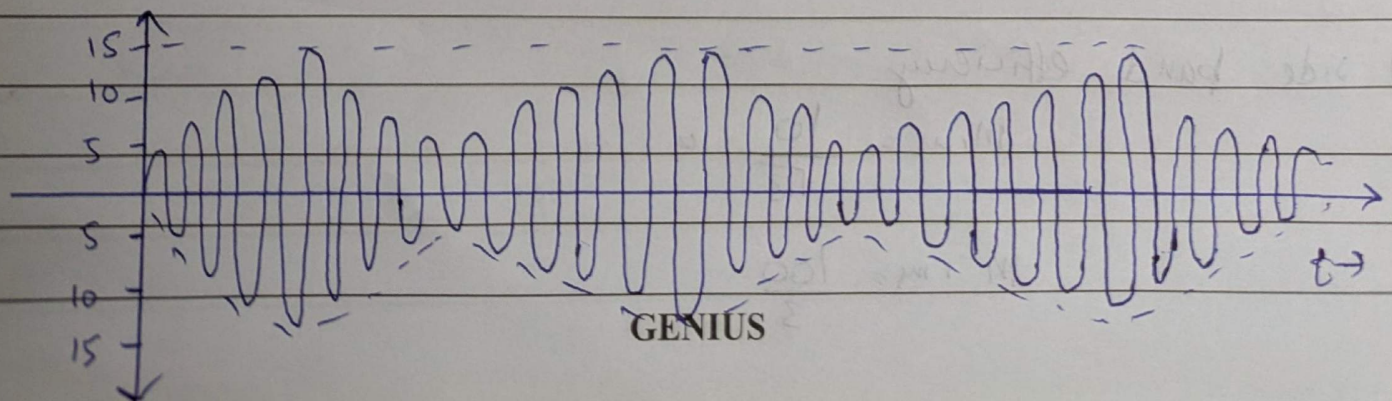
since,

$$-10 \leq m(t) \leq 10$$

The signed ampl is

$$5 + m(t)$$

a) So it varies from -5 to 15 A and its a over modulating AM.



b) The envelope detector will not give the signed quantity directly as, $|A + m(t)|$
it gives the magnitude,

Thus, Output = $|5 + m(t)|$ as it gets negative, the negative part gets flipped positive so detector output is not $m(t)$.

Let $m(t) = -10$

$$5 + m(t) = -5$$

but the envelope detector gives $|-5| = 5$
that is not equal to -10 .

c) given

$$s(t) = A_c [A + m(t)] \cos \omega_c t$$

determining the envelope, we take amplitude factor multiplying the $\cos(\omega_c t)$:

$$E(t) = |A_c [A + m(t)]|$$

since $A_c > 0$

$$\boxed{E(t) = A_c |A + m(t)|}$$
 is the output

4.3-7:

given $\mu = 1$

a) we know that

$$\mu = \frac{m_{\max}}{A_c}$$

$$m_{\max} = 1$$

we get.

$$1 = \frac{1}{A_c}$$

$$\boxed{A_c = 1}$$

$$\text{Power} = \frac{A_c^2}{2} = \boxed{\frac{1}{2} \text{ W}}$$

b)

AM signal $s(t) = [1 + m(t)] \cos \omega_c t$ since $m(t) = \pm 1$

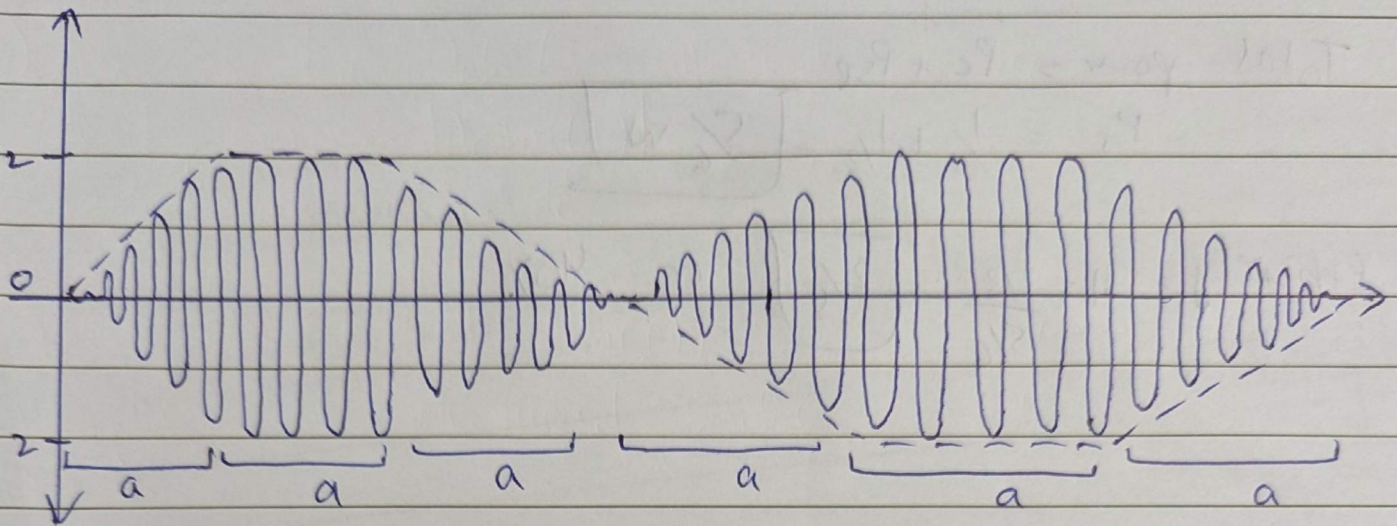
$$1 + (-1) = 0$$

$$1 + (1) = 2$$

So,

$$0 \leq \text{envelope} \leq 2$$

because $\mu = 1$ so it never goes negative.



c) rising ramp $-1 \rightarrow +1$
top $+1$

falling ramp $+1 \rightarrow -1$

bottom -1

and

$$4a = 10^{-3} \text{ s}$$

for linear ramp:

$$(m^2)_{\text{ramp}} = \frac{1}{3}$$

for flat section

$$m^2 = 1$$

$$E[m^2] = \frac{1}{4} \left(\frac{1}{3} + 1 + \frac{1}{3} + 1 \right) \quad \because \text{all 4 sections have equal duration.}$$

$$E[m^2] = \frac{2}{3}$$

$$\text{Side band power} = \frac{1}{2} E[m^2] = \boxed{\frac{1}{3} W}$$

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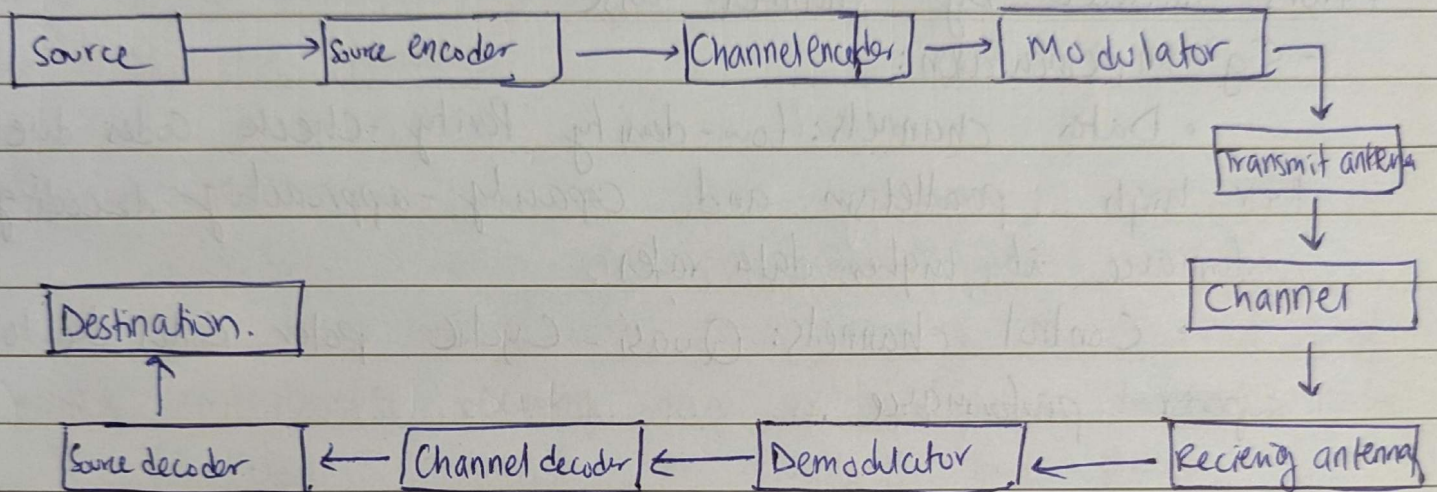
Total power = $P_c + P_{se}$

$$P_T = \frac{1}{2} + \frac{1}{3} = \boxed{\frac{5}{6} \text{ W}}$$

$$\text{efficiency} = \eta = \frac{\frac{1}{3}}{\frac{5}{6}} = \boxed{\frac{2}{5}} \text{ or } 40\%$$

Analog and digital communication Part - II

• System level analysis of 5G New Radio:



- Source: generates raw info, modern networks consist of continuous ~~signal~~ analog signals or discrete digital data sets. It converts raw data into uncompressed digital bitstreams (b/s) via high resolution ADCs.
- Source encoder: It removes statistical redundancy to minimize the total required bit rate before transmission.
for audio: enhanced voice services codec operating across bitrates from 5.9 kbps to 128 kbps.
for video: high efficiency coding or versatile video coding to compress 4K/8K media streams.

• Channel encoder:

Introduce controlled parity bits to the compressed bit stream, allowing the receiver to detect and correct bit errors induced by channel noise.

- 5g implementation:

- Data channels: Low-density Parity-check codes due to their high parallelism and capacity-approaching decoding performance at higher data rates.

- Control channels: Quasi-Cyclic polar codes due to superior performance.

• Modulator:

It maps ~~the~~ ~~the~~ ~~the~~ ~~the~~ ~~the~~ digital binary sequences into complex baseband symbols and converts them into radio frequency carrier waveforms.

- Maps bits into multi-level constellation points and applies Cyclic Prefix orthogonal frequency division multiplexing.

• Transmit antenna:

Converts conducted high-frequency AC electrical signal into unguided electromagnetic wave radiated into free space. It uses Active Antenna Units incorporating Massive MIMO arrays with digital beamforming to direct narrow transmission beams precisely towards active devices.

- Channel:

The physical medium through which waves propagate. It introduces path loss, shadowing, Doppler shifting from mobility and severe multipath fading caused by reflections from buildings and terrain.

- Receiver antenna:

Catches incoming electromagnetic radiations from space and turn them back into high-frequency AC electrical signal.

- uses multiplexing recovery using multi-element antenna arrays at the user equipment or gNodeB base station.

- De modulator:

Performs signal downconversion, timing/frequency ~~set~~ synchronization, channel equalization and symbol demapping. It removes cyclic prefix, executes (FFT) routines to extract subcarrier values, calculates soft-log-likelihood Ratios for each bit.

- Channel decoder:

uses LLR values and injected parity constraints to locate and resolve bit inversions caused by channel corruption. executes belief propagation (BP) for LDPC data blocks, or Successive cancellation list decoding for Polar control payloads.

- Source decoder:

Reverses some compression algorithm to reconstruct the original digital representation of the transmitted message.

- Destination:

Converts reconstructed digital information back into physical human-perceivable energy forms or machine-processable data.
ex: Displays, speakers, or actuators, etc.

- Modern Techniques in 5G New Radio:

1. Data rate maximization techniques:

- MIMO & Spatial Multiplexing:

Base station utilizes large-scale antenna array to transmit upto 16 parallel, non-interleaved data streams simultaneously over the exact same frequency band via precise spatial channel orthogonalization.

- High-order Modulation: Maps upto 8 bits per symbol or 10 bits per symbol under high SNR channel condition.

2. Latency Reduction technique:

• Flexible Numerology:

Unlike LTE's fixed 15 kHz subcarrier spacing, 5G NR scales subcarrier spacing by $\Delta f = 15 \times 2^M$ kHz ($\therefore M = 1, 2, 3, \dots$).

• Mini-slots & Non-slot Based Scheduling:

enables Ultra-Reliable Low-latency communication (URLLC) traffic to interrupt and preempt ongoing scheduled data instantly.

3- Quality of service (QoS) Techniques:

• Adaptive modulation and Coding:

UEs measure link quality using channel state information reference signals and send periodic channel quality indicator reports.

• Network slicing:

Dynamically partition physical network infrastructure into distinct logical end-to-end networks optimized for specific QoS parameters.